Unit 3

Lesson Two

Methods of integration

1 The integration by substitution :

There are several methods to change integrations that are not on standard form to be in standard forms and the easiest and the most common method is the substitution method , taking in consideration the following :

- If the given integration is given in the form $\int f(g(x)).\, \grave{g}(x).\, dx$ use the substitution $g(x) = z$
- If the given integration contains the nth root of a function as $\sqrt[n]{g(x)}$, use the substitution $g(x) = z^n$ or $g(x) = z$
- In some questions you can use an appropriate substitution in order to simplify the integration and rewrite it in standard form.

Example 1

Find each of the following :

1. $\int x^3 (2x^4 + 3)^5 \, dx$
2. $\int \frac{x+3}{(x^2+6x)^7} \, dx$
3. $\int x^3 (x^2 + 1)^5 \, dx$
4. $\int \sqrt{x} \left(\sqrt{x} - 2\right)^5 \, dx$

Solution

1. Put $z = 2x^4 + 3$ $\quad\therefore dz = 8x^3 \, dx$

$$\therefore \int (2x^4 + 3)^5 \times x^3 \, dx = \frac{1}{8} \int (2x^4 + 3)^5 \times 8x^3 \, dx$$
$$= \frac{1}{8} \int z^5 \, dz = \frac{1}{8} \times \frac{1}{6} z^6 + c$$
$$= \frac{1}{48} (2x^4 + 3)^6 + c$$

2. Put $z = x^2 + 6x$ $\quad\therefore dz = (2x + 6)\, dx = 2(x + 3)\, dx$

$$\therefore \int \frac{x+3}{(x^2+6x)^7} \, dx = \frac{1}{2} \int (x^2 + 6x)^{-7} \times 2(x + 3) \, dx$$
$$= \frac{1}{2} \int z^{-7} \, dz = \frac{-1}{12} z^{-6} + c$$
$$= \frac{-1}{12} (x^2 + 6x)^{-6} + c$$

3. Put $z = x^2 + 1$ $\quad\therefore x^2 = z - 1$

$\therefore x = (z-1)^{\frac{1}{2}}$ $\quad\therefore dx = \frac{1}{2}(z-1)^{\frac{-1}{2}}\,dz$

$$\therefore \int x^3 (x^2+1)^5\,dx = \int (z-1)^{\frac{3}{2}} \times z^5 \times \frac{1}{2}(z-1)^{\frac{-1}{2}}\,dz$$

$$= \frac{1}{2}\int z^5 (z-1)\,dz = \frac{1}{2}\int (z^6 - z^5)\,dz$$

$$= \frac{1}{14}z^7 - \frac{1}{12}z^6 + c$$

$$= \frac{1}{14}(x^2+1)^7 - \frac{1}{12}(x^2+1)^6 + c$$

4. Put $z = \sqrt{x} - 2$ $\quad\therefore \sqrt{x} = z + 2$

$\therefore x = (z+2)^2$ $\quad\therefore dx = 2(z+2)\,dz$

$$\therefore \int \sqrt{x}\left(\sqrt{x}-2\right)^5\,dx = \int (z+2) \times z^5 \times 2(z+2)\,dz$$

$$= 2\int z^5 (z+2)^2\,dz$$

$$= 2\int z^5 (z^2 + 4z + 4)\,dz$$

$$= 2\int (z^7 + 4z^6 + 4z^5)\,dz$$

$$= \frac{1}{4}z^8 + \frac{8}{7}z^7 + \frac{4}{3}z^6 + c$$

$$= \frac{1}{4}\left(\sqrt{x}-2\right)^8 + \frac{8}{7}\left(\sqrt{x}-2\right)^7 + \frac{4}{3}\left(\sqrt{x}-2\right)^6 + c$$

Example 2

Find each of the following :

1. $\int x^2 \sqrt[3]{2x+1}\,dx$

2. $\int x^3 \sqrt{1-x^2}\,dx$

3. $\int \frac{x}{\sqrt{x+2}}\,dx$

Solution

1. Put $z^3 = 2x + 1$ $\quad\therefore 2x = z^3 - 1$ $\quad\therefore x = \frac{1}{2}(z^3 - 1)$

$\therefore dx = \frac{3}{2}z^2\,dz$

$$\therefore \int x^2 \sqrt[3]{2x+1}\,dx = \int \frac{1}{4}(z^3-1)^2 \times z \times \frac{3}{2}z^2\,dz$$

$$= \int \frac{3}{8}z^3 (z^6 - 2z^3 + 1)\,dz$$

$$= \int \left(\frac{3}{8} z^9 - \frac{3}{4} z^6 + \frac{3}{8} z^3\right) dz$$

$$= \frac{3}{80} z^{10} - \frac{3}{28} z^7 + \frac{3}{32} z^4 + c$$

$$= \frac{3}{80} (2x+1)^{\frac{10}{3}} - \frac{3}{28} (2x+1)^{\frac{7}{3}} + \frac{3}{32} (2x+1)^{\frac{4}{3}} + c$$

Another Solution

Put $z = 2x + 1$ $\quad \therefore 2x = z - 1$ $\quad \therefore x = \frac{1}{2}(z-1)$

$\therefore dx = \frac{1}{2} dz$

$$\therefore \int x^2 \sqrt[3]{2x+1}\, dx = \int \frac{1}{4} (z-1)^2 \sqrt[3]{z} \times \frac{1}{2} dz$$

$$= \frac{1}{8} \int (z^2 - 2z + 1) \times z^{\frac{1}{3}} dz$$

$$= \frac{1}{8} \int \left(z^{\frac{7}{3}} - 2z^{\frac{4}{3}} + z^{\frac{1}{3}}\right) dz$$

$$= \frac{3}{80} z^{\frac{10}{3}} - \frac{3}{28} z^{\frac{7}{3}} + \frac{3}{32} z^{\frac{4}{3}} + c$$

$$= \frac{3}{80} (2x+1)^{\frac{10}{3}} - \frac{3}{28} (2x+1)^{\frac{7}{3}} + \frac{3}{32} (2x+1)^{\frac{4}{3}} + c$$

2. Put $z = 1 - x^2$ $\quad \therefore x^2 = 1 - z$ $\quad \therefore x = (1-z)^{\frac{1}{2}}$

$\therefore dx = \frac{-1}{2} (1-z)^{\frac{-1}{2}} dz$

$$\therefore \int x^3 \sqrt{1-x^2}\, dx = \int (1-z)^{\frac{3}{2}} \times \sqrt{z} \times \frac{-1}{2} (1-z)^{\frac{-1}{2}} dz$$

$$= \frac{-1}{2} \int z^{\frac{1}{2}} (1-z)\, dz$$

$$= \frac{-1}{2} \int \left(z^{\frac{1}{2}} - z^{\frac{3}{2}}\right) dz$$

$$= \frac{-1}{2} \left[\frac{2}{3} z^{\frac{3}{2}} - \frac{2}{5} z^{\frac{5}{2}}\right] + c$$

$$= -\frac{1}{3} z^{\frac{3}{2}} + \frac{1}{5} z^{\frac{5}{2}} + c$$

$$= \frac{-1}{3} (1-x^2)^{\frac{3}{2}} + \frac{1}{5} (1-x^2)^{\frac{5}{2}} + c$$

3. Put $z = x + 2$ $\quad \therefore x = z - 2$ $\quad \therefore dx = dz$

$$\therefore \int \frac{x}{\sqrt{x+2}}\, dx = \int \frac{z-2}{\sqrt{z}}\, dz = \int \left(z^{\frac{1}{2}} - 2z^{\frac{-1}{2}}\right) dz$$

$$= \frac{2}{3} z^{\frac{3}{2}} - 4z^{\frac{1}{2}} + c$$

$$= \frac{2}{3} (x+2)^{\frac{3}{2}} - 4(x+2)^{\frac{1}{2}} + c$$

Remarks

1. $\int (f(x))^n \grave{f}(x)\, dx = \frac{(f(x))^{n+1}}{n+1} + c$

   For example : $\int (x^5+2)^7 \times 5x^4\, dx = \frac{(x^5+2)^8}{8} + c$

2. $\int \frac{\grave{f}(x)}{f(x)} dx = \ln|f(x)| + c$

   "If the numerator is derivative of the denominator , then the integration = ln | the denominator | + c"

   For example : $\int \frac{3x^2+2x}{x^3+x^2+1} dx = \ln|x^3+x^2+1| + c$

3. $\int e^{f(x)} \grave{f}(x)\, dx = e^{f(x)} + c$

   For example : $\int e^{7x^2+3} \times 14x\, dx = e^{7x^2+3} + c$

4. $\int \frac{\grave{f}(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + c$

   "If the numerator is derivative of what inside the root in the denominator , then the integration = 2 the root + c"

   For example : $\int \frac{2x}{\sqrt{x^2+15}} dx = 2\sqrt{x^2+15} + c$

We can use the integration by substitution to prove the previous remarks by putting $z = f(x)$

$\therefore dz = \grave{f}(x)\, dx$

i.e.

1. $\int (f(x))^n \grave{f}(x)\, dx = \int z^n\, dz = \frac{z^{n+1}}{n+1} + c = \frac{[f(x)]^{n+1}}{n+1} + c$

2. $\int \frac{\grave{f}(x)}{f(x)} dx = \int \frac{dz}{z} = \ln|z| + c = \ln|f(x)| + c$

3. $\int e^{f(x)} \grave{f}(x)\, dx = \int e^z\, dz = e^z + c = e^{f(x)} + c$

4. $\int \frac{\grave{f}(x)}{\sqrt{f(x)}} dx = \int \frac{dz}{\sqrt{z}} = 2\sqrt{z} + c = 2\sqrt{f(x)} + c$

We can use the remarks in the solution of the problems directly without doing the substitution process.

Example 3

Find each of the following :

1. $\int m\sqrt{m^2-1}\,dm$

2. $\int \frac{x^2+1}{\sqrt{x^3+3x-2}}\,dx$

3. $\int \frac{2x-3}{x^2-3x}\,dx$

4. $\int x e^{x^2}\,dx$

Solution

1. $\int m\sqrt{m^2-1}\,dm = \frac{1}{2}\int (m^2-1)^{\frac{1}{2}} \times 2m\,dm$

"a function is multiplied by its derivative".

$$= \frac{1}{3}(m^2-1)^{\frac{3}{2}} + c$$

2. $\int \frac{x^2+1}{\sqrt{x^3+3x-2}}\,dx = \frac{1}{3}\int \frac{3x^2+3}{\sqrt{x^3+3x-2}}\,dx = \frac{1}{3} \times 2\sqrt{x^3+3x-2} + c$

$$= \frac{2}{3}\sqrt{x^3+3x-2} + c$$

" The numerator is derivative of what inside the root in the denominator ".

3. $\int \frac{2x-3}{x^2-3x}\,dx = \ln|x^2-3x| + c$ " The numerator is derivative of the denominator ".

4. $\int x e^{x^2}\,dx = \frac{1}{2}\int e^{x^2} \times 2x\,dx = \frac{1}{2}e^{x^2} + c$

Example 4

Find each of the following :

1. $\int x(x+2)^8\,dx$

2. $\int \sqrt[5]{2-\sqrt{x}}\,dx$

3. $\int \sqrt{\frac{1+\sqrt{x}}{x}}\,dx$

4. $\int \frac{\sqrt{\ln x}}{x}\,dx$

Solution

1. Put $z = x+2$ $\quad\therefore x = z-2$ $\quad\therefore dx = dz$

$$\therefore \int x(x+2)^8\,dx = \int (z-2) \times z^8\,dz = \int (z^9 - 2z^8)\,dz$$

$$= \frac{1}{10}z^{10} - \frac{2}{9}z^9 + c$$

$$= \frac{1}{10}(x+2)^{10} - \frac{2}{9}(x+2)^9 + c$$

Another Solution

$$\int x(x+2)^8\,dx = \int (x+2-2)(x+2)^8\,dx$$

$$= \int \left[(x+2)^9 - 2(x+2)^8\right] dx$$

$$= \frac{1}{10}(x+2)^{10} - \frac{2}{9}(x+2)^9 + c$$

Notice that

We can rewrite $x = (x+2-2)$ and continue the solution without the substitution.

2. Put $z^5 = 2 - \sqrt{x}$ $\quad\therefore \sqrt{x} = 2 - z^5$

$\therefore x = (2-z^5)^2 = 4 - 4z^5 + z^{10}$

$\therefore dx = (-20z^4 + 10z^9)\,dz$

$$\therefore \int \sqrt[5]{2-\sqrt{x}}\,dx = \int z(-20z^4 + 10z^9)\,dz$$

$$= \int (-20z^5 + 10z^{10})\,dz = \frac{-20}{6}z^6 + \frac{10}{11}z^{11} + c$$

$$= \frac{-10}{3}\left(\sqrt[5]{2-\sqrt{x}}\right)^6 + \frac{10}{11}\left(\sqrt[5]{2-\sqrt{x}}\right)^{11} + c$$

3. Put $z = 1 + \sqrt{x}$ $\quad\therefore \sqrt{x} = z - 1$

$\therefore x = (z-1)^2$ $\quad\therefore dx = 2(z-1)\,dz$

$$\therefore \int \sqrt{\frac{1+\sqrt{x}}{x}}\,dx = \int \sqrt{\frac{z}{(z-1)^2}} \times 2(z-1)\,dz = 2\int \sqrt{z}\,dz$$

$$= 2\int z^{\frac{1}{2}}\,dz = \frac{4}{3}z^{\frac{3}{2}} + c = \frac{4}{3}\left(1+\sqrt{x}\right)^{\frac{3}{2}} + c$$

4. Put $\ln x = z$ $\quad\therefore \frac{1}{x}dx = dz$

$$\therefore \int \frac{\sqrt{\ln x}}{x}\,dx = \int \sqrt{z}\,dz = \int z^{\frac{1}{2}}\,dz = \frac{2}{3}z^{\frac{3}{2}} + c$$

$$= \frac{2}{3}(\ln x)^{\frac{3}{2}} + c$$

Example 5

Find each of the following :

1. $\int \cos^5 x \sin x\,dx$

2. $\int \cot x \csc^3 x\,dx$

3. $\int \sqrt{1+\sin x}\cos x\,dx$

4. $\int (\sin x + \cot x)^8 (\cos x - \csc^2 x)\,dx$

5. $\int x^2 \sec^2(x^3+5)\,dx$

6. $\int \csc^2 x\, e^{\cot x}\,dx$

Solution

1. Put $z = \cos x$

$\therefore dz = -\sin x\, dx$

$\therefore \int \cos^5 x \sin x\, dx = -\int z^5\, dz = -\frac{1}{6} z^6 + c$

$= -\frac{1}{6} \cos^6 x + c$

2. Put $z = \csc x$ $\quad\quad$ $\therefore dz = -\csc x \cot x\, dx$

$\therefore \int \cot x \csc^3 x\, dx = \int \csc^2 x \times \csc x \cot x\, dx$

$= -\int z^2\, dz = -\frac{1}{3} z^3 + c = -\frac{1}{3} \csc^3 x + c$

3. Put $z = 1 + \sin x$ $\quad\quad$ $\therefore dz = \cos x\, dx$

$\therefore \int \sqrt{1+\sin x} \cos x\, dx = \int z^{\frac{1}{2}} dz = \frac{2}{3} z^{\frac{3}{2}} + c = \frac{2}{3} (1+\sin x)^{\frac{3}{2}} + c$

4. Put $z = \sin x + \cot x$ $\quad\quad$ $\therefore dz = (\cos x - \csc^2 x)\, dx$

$\therefore \int (\sin x + \cot x)^8 (\cos x - \csc^2 x)\, dx = \int z^8\, dz = \frac{1}{9} z^9 + c$

$= \frac{1}{9} (\sin x + \cot x)^9 + c$

5. Put $z = x^3 + 5$ $\quad\quad$ $\therefore dz = 3x^2\, dx$

$\therefore \int x^2 \sec^2 (x^3+5)\, dx = \frac{1}{3} \int \sec^2 (x^3+5) \times 3x^2\, dx$

$= \frac{1}{3} \int \sec^2 z\, dz = \frac{1}{3} \tan z + c = \frac{1}{3} \tan (x^3+5) + c$

6. Put $z = \cot x$ $\quad\quad$ $\therefore dz = -\csc^2 x\, dx$

$\therefore \int \csc^2 x\, e^{\cot x}\, dx = -\int e^z\, dz = -e^z + c = -e^{\cot x} + c$

Remark

We can solve this example by substitution method or by using this rule :

$\int [f(x)]^n \times f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$

, where $n \neq -1$

2 Integration by parts

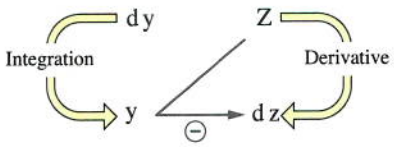
If y and z are two functions in the variable x and differentiable , then

$\frac{d}{dx}(yz) = y\frac{dz}{dx} + z\frac{dy}{dx}$

By integrating both sides with respect to x

$\therefore \int \frac{d}{dx}(yz)\, dx = \int y\frac{dz}{dx}\, dx + \int z\frac{dy}{dx}\, dx$

$\therefore yz = \int y\, dz + \int z\, dy$ $\quad\quad$ $\therefore \int z\, dy = yz - \int y\, dz$



i.e. $\int$ (The product of two functions) = (1st function) × (integrating the 2nd) $-\int$ (integrating the 2nd) × (derivative of 1st)

and this equation is called the rule of the integration by parts which is used to find the integration of the product of two function one of them is not the derivative of the other by using the suitable choosing of each of y and z and the following examples illustrate that.

Example 6

Find each of the following :

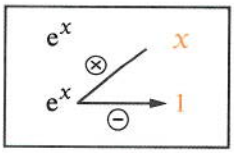
1. $\int x\,e^{x}\,dx$ **2.** $\int \frac{x}{e^{3x}}\,dx$

Solution

1. $\int x\,e^{x}\,dx = x\,e^{x} - \int e^{x}\,dx$

$= x\,e^{x} - e^{x} + c$

$= e^{x}(x-1) + c$



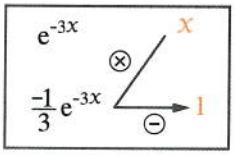
2. $\int \frac{x}{e^{3x}}\,dx = \int x\,e^{-3x}\,dx$

$= \frac{-1}{3}x\,e^{-3x} - \int -\frac{1}{3}e^{-3x}\,dx$

$= \frac{-1}{3}x\,e^{-3x} + \frac{1}{3}\int e^{-3x}\,dx$

$= \frac{-1}{3}x\,e^{-3x} + \frac{1}{3}\times\frac{-1}{3}e^{-3x} + c$

$= \frac{-1}{3}x\,e^{-3x} - \frac{1}{9}e^{-3x} + c$



Notice that

If the function is a product of "an exponential function" by "a polynomial function" , then the polynomial is which to be derivative.

Example 7

Find each of the following :

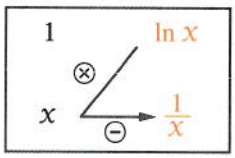
1. $\int \ln x\,dx$ **2.** $\int x^{2}\ln x\,dx$ **3.** $\int \left(\frac{\ln x}{\sqrt{x}}\right)dx$

Solution

1. $\int \ln x\,dx = x\ln x - \int \frac{1}{x}\times x\,dx$

$= x\ln x - \int dx$

$= x\ln x - x + c$

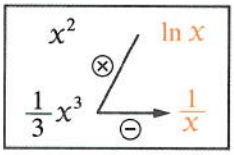


2. $\int x^{2}\ln x\,dx = \frac{1}{3}x^{3}\ln x - \int \frac{1}{3}x^{3}\times\frac{1}{x}\,dx$

$= \frac{1}{3}x^{3}\ln x - \frac{1}{3}\int x^{2}\,dx$

$= \frac{1}{3}x^{3}\ln x - \frac{1}{9}x^{3} + c$

$= \frac{1}{9}x^{3}[3\ln x - 1] + c$



Notice that

If the function is a product of "a logarithmic -exponential function" by "a polynomial function" , then the "logarithmic -exponential" is which to be derivative.

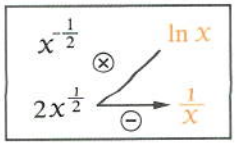
3. $\int \left(\frac{\ln x}{\sqrt{x}}\right) dx = \int (\ln x)\left(x^{\frac{-1}{2}}\right) dx$

$= 2x^{\frac{1}{2}} \ln x - \int \frac{2}{x} \times x^{\frac{1}{2}} dx$

$= 2x^{\frac{1}{2}} \ln x - 2\int x^{\frac{-1}{2}} dx$

$= 2x^{\frac{1}{2}} \ln x - 4x^{\frac{1}{2}} + c$

$= 2\sqrt{x}\,[\ln x - 2] + c$



Example 8

Find each of the following :

1. $\int x^2 e^x dx$ **2.** $\int (x-1)^2 e^{2x} dx$

Solution

1. $\int x^2 e^x dx$

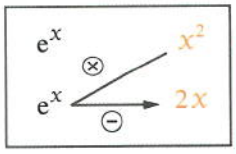
$= x^2 e^x - 2\int x e^x dx$

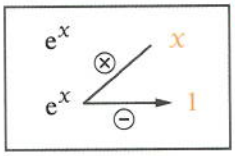
$= x^2 e^x - 2\left[x e^x - \int e^x dx\right]$

$= x^2 e^x - 2x e^x + 2\int e^x dx$

$= x^2 e^x - 2x e^x + 2e^x + c$

$= e^x (x^2 - 2x + 2) + c$





Notice that

We can do the integration by parts more than one.

Another Solution

By using column of (derivative - integration)

* Put x^2 in derivative column and derivative it untill ends the column by zero.
* Put e^x in integration column and integration it by the same numbers of derivative times.
* The result is the sum of the product of the diagonals by alternating signs as in opposite table.

Integration	derivative
e^x	x^2 (⊕)
e^x	$2x$ (⊖)
e^x	2 (⊕)
e^x	zero

$\therefore \int x^2 e^x dx = x^2 e^x - 2x e^x + 2e^x + c$

2. $\int (x-1)^2 e^{2x} dx$

$= \frac{1}{2}(x-1)^2 e^{2x} - \frac{1}{2}(x-1) e^{2x} + \frac{1}{4} e^{2x} + c$

Integration	derivative
e^{2x}	$(x-1)^2$ (⊕)
$\frac{1}{2}e^{2x}$	$2(x-1)$ (⊖)
$\frac{1}{4}e^{2x}$	2 (⊕)
$\frac{1}{8}e^{2x}$	zero

Example 9

Find each of the following :

1. $\int x^2 e^{x^3} dx$ **2.** $\int x^3 e^{x^2} dx$

Solution

1. Put $z = x^3$ $\therefore dz = 3x^2 dx$

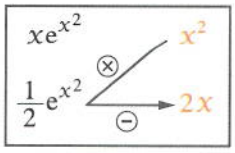
$$\therefore \int x^2 e^{x^3} dx = \frac{1}{3}\int e^{x^3}(3x^2 dx)$$
$$= \frac{1}{3}\int e^z dz = \frac{1}{3}e^z + c$$
$$= \frac{1}{3}e^{x^3} + c$$

Notice that

In the 1st problem we use the substitution method but the 2nd problem solved by integrating by parts.

2. Factorize $x^3 e^{x^2}$ into $x^2 \times x e^{x^2}$

$$\therefore \int x^3 e^{x^2} dx = \frac{1}{2}x^2 e^{x^2} - \int x e^{x^2} dx$$
$$= \frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2} + c$$
$$= \frac{1}{2}e^{x^2}(x^2 - 1) + c$$



Example 10

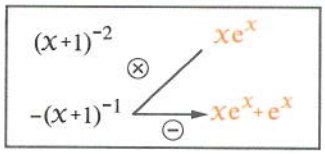
Find each of the following :

1. $\int \frac{x e^x}{(x+1)^2} dx$ **2.** $\int (1 + 4x^4) e^{x^4} dx$

Solution

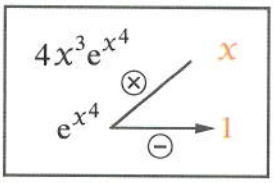
1. $\int \frac{x e^x}{(x+1)^2} dx = \int x e^x (x+1)^{-2} dx$

$$= -x e^x (x+1)^{-1} + \int (x+1)^{-1}(x e^x + e^x) dx$$
$$= \frac{-x e^x}{(x+1)} + \int \frac{e^x (x+1)}{(x+1)} dx = \frac{-x e^x}{x+1} + e^x + c$$



2. $\int (1 + 4x^4) e^{x^4} dx = \int e^{x^4} dx + \int 4x^4 e^{x^4} dx$

$$= \int e^{x^4} dx + \int x \times 4x^3 e^{x^4} dx$$
$$= \int e^{x^4} dx + x\int e^{x^4} - \int e^{x^4} dx$$
$$= x e^{x^4} + c$$



Notice that

$\int f(x) dx - \int f(x) dx$ = constant (not necessary zero)

For example : $\int x^2 dx - \int x^2 dx = \left(\frac{1}{3}x^3 + c_1\right) - \left(\frac{1}{3}x^3 + c_2\right) = c_1 - c_2 = \text{constant}.$

Example 11

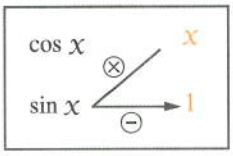
Find each of the following :

1. $\int x \cos x \, dx$ **2.** $\int x \cos x^2 \, dx$

Solution

1. $\int x \cos x \, dx = x \sin x - \int \sin x \, dx$

$= x \sin x + \cos x + c$



2. Put $z = x^2$

$\therefore dz = 2x \, dx$

$\therefore \int x \cos x^2 \, dx = \frac{1}{2} \int 2x \cos x^2 \, dx$

$= \frac{1}{2} \int \cos z \, dz$

$= \frac{1}{2} \sin z + c = \frac{1}{2} \sin x^2 + c$

Notice that

In (1) the solution by using integration by parts method but in (2) the solution by using integration by substituting method.

Remark

There are some integrals if it is calculated by parts twice , then the original integral appears in the second time , then we can calculate it as in the following example.

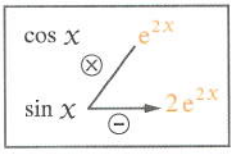
Example 12

Find $\int e^{2x} \cos x \, dx$

Solution

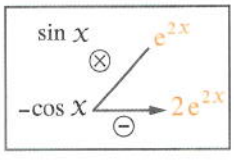
$\because$ The integral (K) $= \int e^{2x} \cos x \, dx$

$= e^{2x} \sin x - 2 \int e^{2x} \sin x \, dx$



$\therefore k = e^{2x} \sin x - 2 [- e^{2x} \cos x + 2 \int e^{2x} \cos x \, dx]$

$= e^{2x} \sin x + 2 e^{2x} \cos x - 4 \int e^{2x} \cos x \, dx$



$\therefore k = e^{2x} \sin x + 2 e^{2x} \cos x - 4k$

by adding 4 k to both sides

$\therefore 5k = e^{2x} \sin x + 2 e^{2x} \cos x + c$

$\therefore k = \frac{1}{5} e^{2x} [\sin x + 2 \cos x] + c$

Notice that

The difference between two identically integrals equals constant and is not necessary to be zero

Enrich your knowledge

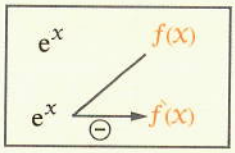
$$\int e^{x}[f'(x)+f(x)]\,dx = e^{x} f(x) + c$$

For example :

$$\int e^{x}[\sec^2 x + \tan x]\,dx = e^{x}\tan x + c$$

Proof :

$$\int e^{x}[f'(x)+f(x)]\,dx = \int e^{x} f'(x)\,dx + \int e^{x} f(x)\,dx$$
$$= \int e^{x} f'(x)\,dx + e^{x} f(x) - \int e^{x} f'(x)\,dx$$
$$= e^{x} f(x) + c$$



By the same way we can prove that

$$\int e^{-x}[f'(x)-f(x)]\,dx = e^{-x} f(x) + c$$

* Notice that we cannot use «the enrich your konwledge» as a rule at the solution but we use the idea of its proof in the solution as the following example.

Example 13

Find each of the following :

1. $\int e^{x}(\cos x + \sin x)\,dx$

2. $\int e^{x}(\cot x - \csc^2 x)\,dx$

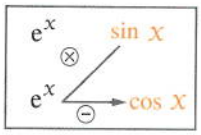
Solution

1. $\int e^{x}(\cos x + \sin x)\,dx$

$= \int e^{x}\cos x\,dx + \int e^{x}\sin x\,dx$

$= \int e^{x}\cos x\,dx + e^{x}\sin x - \int e^{x}\cos x\,dx$

$= e^{x}\sin x + c$

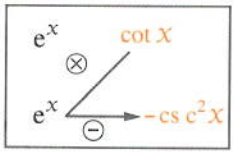


2. $\int e^{x}(\cot x - \csc^2 x)\,dx$

$= \int e^{x}\cot x\,dx - \int e^{x}\csc^2 x\,dx$

$= e^{x}\cot x + \int e^{x}\csc^2 x\,dx - \int e^{x}\csc^2 x\,dx$

$= e^{x}\cot x + c$



Example 14

Find by two different methods : $\int \frac{6x}{\sqrt[3]{2x+3}}\,dx$

Solution

1. Solution by substitution method :

Let $z = 2x + 3$ $\quad\therefore 2x = z - 3$

$\therefore x = \frac{1}{2}(z-3)$ $\quad\therefore dx = \frac{1}{2}\,dz$

$$\therefore \int \frac{6x}{\sqrt[3]{2x+3}}\,dx = \int \frac{3(z-3)}{\sqrt[3]{z}} \times \frac{1}{2}\,dz = \frac{3}{2}\int \left(z^{\frac{2}{3}} - 3z^{\frac{-1}{3}}\right) dz$$

$$= \frac{3}{2}\left(\frac{3}{5}z^{\frac{5}{3}} - \frac{9}{2}z^{\frac{2}{3}}\right) + c = \frac{9}{10}z^{\frac{5}{3}} - \frac{27}{4}z^{\frac{2}{3}} + c = \frac{9}{20}z^{\frac{2}{3}}(2z - 15) + c$$

$$= \frac{9}{20}(2x+3)^{\frac{2}{3}}[2(2x+3) - 15] + c = \frac{9}{20}(2x+3)^{\frac{2}{3}}(4x - 9) + c$$

2. Solution by parts method :

$$\int \frac{6x}{\sqrt[3]{2x+3}}\,dx = \int 6x(2x+3)^{\frac{-1}{3}}\,dx$$

$$= 6x \times \frac{3}{4}(2x+3)^{\frac{2}{3}} - \int 6 \times \frac{3}{4}(2x+3)^{\frac{2}{3}}\,dx$$

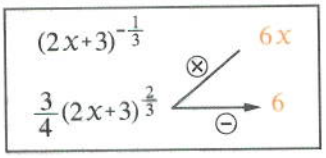
$$= \frac{9}{2}x(2x+3)^{\frac{2}{3}} - \frac{9}{2}\int (2x+3)^{\frac{2}{3}}\,dx$$

$$= \frac{9}{2}x(2x+3)^{\frac{2}{3}} - \frac{9}{2} \times \frac{(2x+3)^{\frac{5}{3}}}{\frac{5}{3}\times 2} + c$$

$$= \frac{9}{2}x(2x+3)^{\frac{2}{3}} - \frac{27}{20}(2x+3)^{\frac{5}{3}} + c$$

$$= \frac{9}{20}(2x+3)^{\frac{2}{3}}\left(10x - 3(2x+3)\right) + c$$

$$= \frac{9}{20}(2x+3)^{\frac{2}{3}}(4x - 9) + c$$



Notice the differences among the methods of solution of each of the following integrations

1. $\int \frac{1}{x \ln x} \, dx$ **2.** $\int \frac{\ln x}{x} \, dx$ **3.** $\int x \ln x \, dx$

Solution

1. $\int \frac{1}{x \ln x} \, dx = \int \frac{\frac{1}{x}}{\ln x} \, dx$ "The numerator is derivative of the denominator"

$= \ln | \ln x | + c$

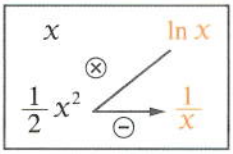
2. $\int \frac{\ln x}{x} \, dx = \int (\ln x) \times \left(\frac{1}{x}\right) dx$ "A function is multiplied by its derivative"

$= \frac{(\ln x)^2}{2} + c$

3. $\int x \ln x \, dx = \frac{1}{2} x^2 \ln x - \int \frac{1}{x} \times \frac{1}{2} x^2 \, dx$ "Integration by parts"

$= \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x \, dx$

$= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + c$



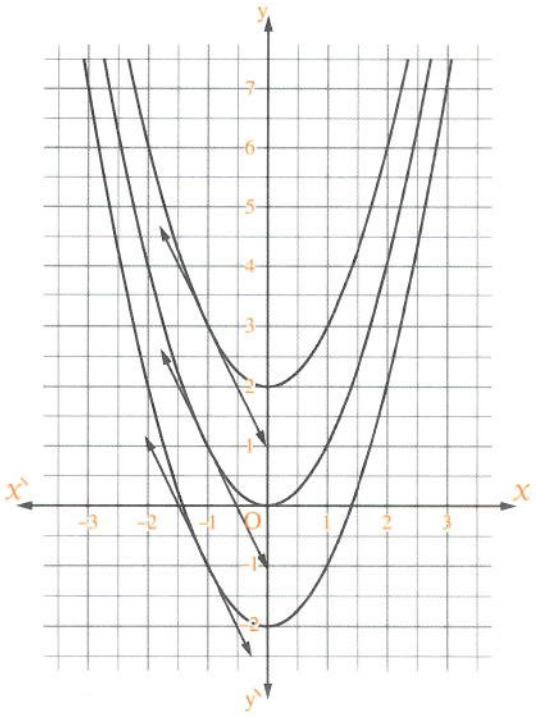
Some applications of the indefinite integration

We know that , if $y = f(x)$ is the equation of a curve , then the slope of the tangent at any point (x , y) on the curve is $\grave{y} = \grave{f}(x)$, and hence the equation of this curve is :

$y = \int \grave{y} \, dx$ or $f(x) = \int \grave{f}(x) \, dx$

Also we know that this integration contains a constant. This constant can take any real value , and hence we get the equation of the family of curves , which has the same tangent at any value of the x-coordinate.

To get one of these family of curves , we must get another condition , say it passes through a given point or cuts a fixed length from one of the axis and so on and the following examples illustrate that.



Example 15

If the slope of the tangent to the curve of the function f at any point (x, y) lying on it is given by the relation $g(x) = x e^{-x}$, find the equation of the curve if it passes through the point $(-1, 3)$

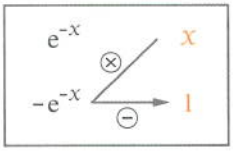
Solution

$\because$ The solpe of the tangent $= x e^{-x}$

$\therefore f(x) = \int x e^{-x} dx$

$= -x e^{-x} - \int -e^{-x} dx = -x e^{-x} - e^{-x} + c$

$= -e^{-x}(x + 1) + c$



, $\because f(-1) = 3$

$\therefore 3 = -e(-1 + 1) + c$ $\quad\quad$ $\therefore c = 3$

$\therefore f(x) = -e^{-x}(x + 1) + 3$

Example 16

If the slope of the normal to the curve of the function f at any point (x, y) lying on it is $\frac{-1}{x\sqrt{x-1}}$, find the equation of the curve if the curve passes through the point $(1, 7)$

Solution

$\because$ The slope of the normal $= \frac{-1}{x\sqrt{x-1}}$

$\therefore$ The slope of the tangent $= x\sqrt{x-1}$ $\quad\quad$ $\therefore f(x) = \int x\sqrt{x-1}\, dx$

, let $z^2 = x - 1$ $\quad\quad$ $\therefore x = z^2 + 1$ $\quad\quad$ $\therefore dx = 2z\, dz$

$\therefore f(x) = \int (z^2 + 1) \times z \times 2z\, dz = \int 2z^2 (z^2 + 1)\, dz$

$= \int (2z^4 + 2z^2)\, dz = \frac{2}{5} z^5 + \frac{2}{3} z^3 + c$

$= \frac{2}{15} z^3 (3z^2 + 5) + c$

$\therefore f(x) = \frac{2}{15} (x - 1)^{\frac{3}{2}} [3(x - 1) + 5] + c = \frac{2}{15} (x - 1)^{\frac{3}{2}} (3x + 2) + c$

, $\because f(1) = 7$

$\therefore 7 = \frac{2}{15} (1 - 1)^{\frac{3}{2}} (3 \times 1 + 2) + c$ $\quad\quad$ $\therefore c = 7$

$\therefore f(x) = \frac{2}{15} (x - 1)^{\frac{3}{2}} (3x + 2) + 7$

Example 17

If the slope of the tangent at any point (X, y) on the curve $y = f(X)$ is $6X^2 - 30X + 36$, then find the equation of this curve, knowing that the value of y at the point of the local maximum equals 28, find also the point of local minimum.

Solution

$\because \grave{f}(X) = 6X^2 - 30X + 36$

$\therefore \grave{f}(X) = 6(X^2 - 5X + 6) = 6(X - 2)(X - 3)$

, $\because \grave{f}(X) = 0$ when $X = 2$, $X = 3$

$\therefore f(X)$ has two critical points at $X = 2$ and $X = 3$

, $\because \acute{f}(X) = 12X - 30$ $\quad\quad$ $\therefore \acute{f}(2) = 24 - 30 = -6 < 0$

$\therefore (2, f(2))$ is a local maximum point.

, $\because \acute{f}(3) = 36 - 30 = 6 > 0$

$\therefore (3, f(3))$ is a local minimum point.

$\because$ The value of y at the point of the local maximum = 28

$\therefore f(2) = 28$

$\therefore (2, 28)$ lies on the curve.

, $\because \grave{f}(X) = 6X^2 - 30X + 36$

$\therefore$ The equation of the family of the curves is

$f(X) = \int \grave{f}(X)\, dX$

i.e. $f(X) = \int (6X^2 - 30X + 36)\, dX$

i.e. $f(X) = 2X^3 - 15X^2 + 36X + c$

, $\because f(2) = 28$

$\therefore 28 = 16 - 60 + 72 + c$ $\quad\quad$ $\therefore c = 0$

$\therefore$ The equation of the required curve is

$y = 2X^3 - 15X^2 + 36X$

, at $X = 3$

$\therefore y = 54 - 135 + 108 = 27$

$\therefore$ The local minimum point is $(3, 27)$

Example 18

Find the equation of the curve : $y = f(x)$ if $\grave{y} = 6x - 4$ and the point $(1, 5)$ is a local minimum point on it.

Solution

$\because \acute{y} = 6x - 4$

$\therefore \grave{y}$ (the slope of the tangent at any point on it) $= \int \acute{y}\, dx = \int (6x - 4)\, dx$

$\therefore \grave{y} = 3x^2 - 4x + c$

$\because (1, 5)$ is a local minimum point of the function.

$\therefore \grave{y} = 0$ when $x = 1$

by substituting in (1) $\quad \therefore$ zero $= 3 - 4 + c \quad \therefore c = 1$

$\therefore \grave{y} = 3x^2 - 4x + 1$

$\therefore$ The equation of the family of the curves is

$y = \int (3x^2 - 4x + 1)\, dx$

i.e. $y = x^3 - 2x^2 + x + \grave{c}$

, $\because (1, 5)$ on the curve of the function.

$\therefore y = 5$ when $x = 1 \quad \therefore 5 = 1 - 2 + 1 + \grave{c} \quad \therefore \grave{c} = 5$

$\therefore$ The equation of the required curve is $y = x^3 - 2x^2 + x + 5$